

ActInf Textbook Group ~ Cohort 7 ~ Session 3 (Modeling & General Discussion)

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all right greetings everyone thank you
for joining so we're in this first up of
the third
weeks for the
cohort it'll be a bit of a grab bag so
we'll see where it goes but a few people
have written some first questions um and
then we'll look at some prior questions
see what else comes up all
right so first I'd be interested in
examples on how to train an active
inference
model what what would anyone say on
this how is training an active inference
model similar or different than
potentially other kinds of
models um hi uh I can say uh so um I was
thinking like um there should be a world
model which would be trained on
the um on minimizing the variation of
free energy uh as a loss
objective so I mean I'm assuming that
can be for for from from an uh like an
RL perspective or something like from
some collected
trajectories that can be trained but but
for the expected free energy uh I'm I'm
not so confident like how H how how that
can be
deployed yeah thank you
Frasier I would just I mean I've only
successfully made maybe like two or
three active inference models
that are like non-toy examples but one
thing that I found particularly helpful

was or or Salient was that you had to really think about the different time scales of things that we're going to be training so you know I guess in in a machine learning context training refers to the updating of parameters of a model so you know and and as was just uh said by is it three I think it's three we need our our model to embody uh the environment in some way so that's going to be parameterized there G to be a bunch of parameters and we need to tweak those parameters in a machine learning setting um that happens at a much slower or at least a slower time scale than the actions that we need to decide to emit at each time point if it's a a discrete time model let's say so there's kind of two questions there's the question of what actions do we emit based upon expected free energy based upon maybe just vfe and then there's the question of okay what how do we also minimize VF with respect to our parameters so those two questions Loom large at least in my mind when I'm thinking about training an active agent cool yeah well there's there's a lot of pieces to to bring out here um one note is that variational basian methods like the variational auto encoder are also trained or updated using variational free energy minimization so the concept of using the evidence lower bound to to um bound and approximate surprise which is equivalent to maximizing model evidence that's not a new modeling trick

from active inference part of the interesting thing though is bringing the perceptual and the action selection process into the unified objective and then using the evidence lower bound on that so so there's like some I mean it it will be cool to kind of pull out from the conversations and papers and get to the vend diagram so that's that's one thing is training a model based upon free energy minimization is something you'll find in a variety of machine learning methods um an interesting difference in terms of of at least like in terms of how chapter six differs uh from some other machine learning Pathways like nowhere in chapter 6 The the recipe for modeling does it say anything

like go find a data set of positive examples and then tune your model so that it recapitulates the maximum performance on the positive data set so it's kind of interesting is like prior data can have a different role in the situation of training an active inference model so sometimes in machine learning you have a you have a basically a priori prior setting of a general or a large topology of the model like here's the architecture of the neural network and the layers or I'm training a 7 billion parameter Transformer architecture so you kind of have a large generic function approximator where the nodes are essentially not intended to have any semantic

meaning and then during the training phase you pump cats and dogs with labeled examples and then you have the data set of cats and dogs and then you have some held out data set of cats and dogs and you try to improve performance on that so in in this situ in the machine learning situation you you have a kind of generic model architecture usually with many many parameters like thousands to billions and then positive examples are pushed through that

Network so that the weights come to embody patterns in in the data in contrast in at least in terms of how chapter 6 lays out the process the basian graphs that are constructed in active inference are smaller and usually more interpretable where the the nodes have distributions that are kind of like based upon how the problem was carved out to be like you have a node that's explicitly based upon the cat or the dog um so that's kind of a difference is rather than walking into the situation with a sort of

General highly parameter uh uh architecture and a data set here as chapter 6 lays out is you have a bit more of like a systems modeling approach so that that's one important difference but how how do you um update the the parameters

um that's pretty General because there's a lot of different parameters to train Let's uh look at

so coming to like chapter 7 this is just focusing on the downstairs sense making piece of what later action will get plugged into so here prior D hidden State s Through Time State transitions B each state is emitting observations you can think of that as being like the hidden state multiplied by the a ambiguity Matrix leads to an observation distribution that's the generative direction or the the the synthetic data generating direction or you could take an observation multiply it by the ambiguity Matrix and kind of run it back to reconstruct an uncertainty distribution over hidden States so that's part part of the flexibility um depending on the situation you might say well we know the OB observations without error like they might be noisy but we measured them without error and then we um we want to learn some other parameter or we know the hidden States through time but we want to infer the the prior so this kind of gets into which variables are fixed and which variables are learnable the way that a distribution is fixed in a basian setting it's like it has a hyper prior so a prior on that distribution that's just a spike like the the observation was seven so just a spike at seven whereas if it was the observation was seven plus or minus one then that would take in a whole distribution so just to conclude and

give a first pass General answer when
you're set up the model if a variable is
entered as fixed and with a kind of
infinite

Precision on it it's not learnable

whereas distributions that have like
actual

breadth those parameter updates can be
updated within that range of the

distribution what what what there

there's a lot of other pieces that was

totally not a complete answer what are

any thoughts or or questions from

there I guess most broadly you have to

choose a model you have to choose a

particular model something like a pomdp

or or Gan mixture or or something uh

whatever it is that discussion applies

to your your choice of model but you do

have to you do have to choose a

model yeah another just well these are

just thinking of broader modeling

considerations so it was brought up H

how do we how do we train on what hasn't

happened like with expected free

energy so let's look at just a first

pass on what the expected free

energy uh calculation entails it's going

to take in a policy prior Π it's a

vector of policies policy the policy

space just like in other control

theoretic formalizations the policy

space is a vector of all the possible

policies policies are all the

combinations of affordances for a given

time Horizon so if we had four

affordances up down left right time

Horizon of two there'd be

16 policies up up up right Etc we're going to take this policy Vector which is the probability distribution of action it sums to one something must happen and we're going to update the policy prior into the policy posterior policies are going to be sharpened according to their score on this unified objective a good score can be gained from having either a high pragmatic value which is to say that the sequence of expected future observations is aligned with preferences that's utility Andor the policy has a high expected Information Gain or epistemic value which is to say that the KL Divergence about beliefs about hidden States through time is diverging because of the observations that are expected to come in so different policies might be valuated to have different pragmatic values and different epistemic values there is this question though how do you balance these two components of the expected free energy because just being just having both of them in a unified objective doesn't mean that there's going to be um an Adaptive finessing of this explore exploit type tradeoff so um first like here at the parametric level here's what that updating or training looks like it looks like changing the amplitude of a variable so

[Music]

um let's see what

chapter okay so here we are in chapter 7

it's a rat and a team as it has a preference for the attractive stimulus so

um check the script to see whether this is written um in terms of log probabilities Etc but a six in the preference Vector here means that the attractive stimulus is considered e to the 6 times more probable

If This Were one and negative one the relative preferences would have all remained the same If This Were 600 you know you can imagine it would be exponentially more so the way that pragmatic value gets scaled is with scaling the amplitude of the C Vector so you you play around with with you could take the C vector and then just like in the code you could make a series of of C Vector Alternatives like multiply by 0.1 multiply by 0.5 .9 1.1 1.5 2 10 so generate this like Suite of C's bringing you closer to on one hand a flat C like if you said multiply this by one millionth so it's it's basically zeroed out that is equivalent to having a flat preference distribution having a flat preference distribution is equivalent so here's PR now pragmatic is on the first term here having a flat preference distribution is equivalent to Infomax on the other hand you could make your C Vector super massive that would make the epistemic component of the expected free energy

negligible which would essentially look like

utility uh maximizing control so one lever that you have is modifying the pragmatic value that's pretty straightforward because you're just doing a vector lengthening or shortening and then the other way that you can rebalance the scales is with epistemic value um if your parameters are not learnable like if they're isn't anep if there isn't an uncertainty distribution so you can imagine increasing the Precision on your variables until they're essentially fixed then epistemic value goes down conversely if you were to expand your uncertainty on the variables epistemic value would go up so whether by combination of widening and narrowing your uncertainty distributions to manipulate epistemic value or stretching or shrinking the C Vector to modify the pragmatic value so those are kind of the two knobs that you have to to to manipulate the relative intensities of these two terms but then how do you really select a given parameterized tradeoff so that's a situation where um like going back to the um Tas so this not not the only way to do this but this is just one one way to do it would we we set up the teamas with a given setting now as with any training the optimal parameterization is going to depend on the niche so if we had an example where the food was always on the left then the optimal parameterization is going to be

different than if it was switching like a metronome or if it was switching with some autocorrelation function so that's why it does depend on how you set up the niche then you could take let's just say we built five C vectors ranging from a kind of like the 0.1 the 0 five the one the two and the 10 and then five different uncertainty distributions that we could choose from so now we have 25 combinations then we could basically simulate with all 25 of those combinations and then say well which of those mice had the highest performance and says okay well it turns out that given this Niche setup having this C Vector look like this and having the uncertainty distribution look like that gave the highest outcome that's again interesting to consider because it wasn't the case that we had like any positive data set examples so it's a really interesting opportunity to Define these models and then train and tune them in ways that don't necessarily rely on specific past data sets but you can use this sort of synthetic simulation and um method to still develop more useful tunings so that's about the e one option there just being to do parameter sweeps over over possibilities Fraser yeah I was just going to recapitulate that it's so like for you to be able to have an active inference AG per se you have it seems like you have to be able to entertain

counterfactual you know possibilities
about what would be the case if you
didn't an action so you have to be able
to perform actions and you have to be
able to evaluate the suitability of
those actions and the way we do that as
with the EFA let's say
um you're you're gonna have to have some
kind of means of of rank ordering your
choices that you make at points in time
um that's that's kind of the the
ontology of of what you do when you do
training let's say of this kind you can
do you can do variational free energy
minimization but you're not necessarily
doing active inference if you're doing
that you kind of have to have this um
counterfactual uh entertaining of of
trajectories into the future it seems
so yeah
okay let's look over an RX and
fur Okay it is figure 1.2 or is it or is
it
2.1 oh it should be 2.1 is 2.1 yeah
sorry oh no it's it was a
blurring the line what table is in what
document
um so like let's just look at training
in the situation of this um this is not
an action model this is just an
observation model so this is like the
listening to music first part of chapter
7 prior beliefs on the frog and the
Apple then likelihood model that's going
to update based upon an observation
here in this case jumping updates to a
posterior belief distribution
now in in spirit slash um in mathematics

this is exactly what happens with
updating the action prior into the
action

posterior here it's just we're talking
about a belief prior into a belief
posterior conditioned upon an
observation but this is conceptually
what happens like let's let's just say
this was I have a 10% chance of turning
left and then a chance of turning right
90% And then I do G and then update that
policy distribution to look more like
this and then sample the most likely one
or I sample from this distribution so
that's the kind of General concept of of
updating a distribution but this one's
not about

action this isn't even an active
inference model this is just basian
inference yeah yeah this is just basan
inference but we can we can say well
what what would be the here there would
be two things that that might be learned
if we were to kind of build this example
out you could learn your prior that
could come from just simply the prior
posterior like from wherever you left
off in the previous cycle you just pick
up there with your prior um another
method for the prior for learning the
prior

is learning by

counting so learning by counting is is
very related to the earn

model so in this situation you have an
urn like a a a jar and there's a
certain number of balls of color one and
color two so initially you don't know

you may not even know how many colors
there are or maybe you've been told that
there's only the two colors
you pull out color one you but you don't
necessarily want to only believe that
there's color one in there so you'd
start with one in each and then you as
you make
observations um you uh
count and then it's like a coin flip
like if you just kept on Flipping that
coin eventually You' get to 100 and 100
and then if you got to a th and a th
that would be a sharper prior around
50/50 whereas in one would be like a
looser prior so that's one way to learn
beliefs um learning a likelihood model
is kind of another topic so it's like
that's
that's part of the challenge sometimes
is the variables with different
semantics like the hidden State
distributions the a
matrix the B Matrix the consequences of
actions the priors um on preferences c
those all have kind of different ways
that you might approach learning
them because there's different
considerations like what will yield the
most
performance um parameter value then
there are considerations like what is a
cognitively
realistic learning
mechanism another consideration would be
to what extent do we want to learn this
variable and then fix it for
runtime or to what extent do we want

this variable to be learnable or
updatable even in runtime
so in this RX and
for
um model of the figure
2.1 accurately reflecting the
textbook the values are all
fixed and
given but I think it will be very useful
for us to with starting with this
example then continuing on
what does it look like when these
distributions are also allowed
to to be learned in different
ways and crucially for an active
inference agent that has to be by means
of action

yes thank you

Dave are any of the implementations of
RX and fur able to take account of a
situation where not just the values of
the variables but what variables there
are need to be changed in order to
succeed you don't where you you you
didn't know that you have to take
temperature into account where uh to to
distinguish between Apples and frogs you
didn't know that you have to hum to
intervene in to your uh your exploration
where is is do you just assume that you
know is that something that we're even
trying to do at all where there's
genuine Discovery where there's no
precedent for what how the model has to
be changed or change the
dimensionality
of not just the number of Dimensions but
what dimensions there are

actually Kus if you have any thoughts
feel free to
add um I don't really have thoughts uh
behind how this would be attempted I
just
know uh in the AR in first sense that's
been a desire to have such a setup uh
for a while as far as I
understand um for example in the case of
the team Ma I know they had I think it
was Magnus codol he had to write
specifically
code and hardc code the structure I
think uh based on his needs and I I I do
understand how he expressed the need at
the
time to have been able to uh you know uh
to rely on uh structured learning to to
have that but instead he had to uh
provide a custom implementation at that
point but I do believe there's an action
in the orx in3 development team to take
care of this structur approach to
learning in the
future yeah that'll be that'll be great
maybe in the worst case you just have to
do something like uh
completely jump out and do genetic
algorithms
something and entirely outside the
Paradigm if you're if you're lucky you
can simply when you discover relevant
dimensions of actual action or
sensation just say oh well that was in
my in the totality of my prior I just
didn't know it none of the values are
set just off uh go back can change your
the dimensionality the priors and then

recalculate but who knows that that
that's what that's where you get in some
really interesting psychology and really
interesting logical techniques thank you
I I do know that at the end end of um
Magnus kol oh sorry uh Dimitri bv's PhD
thesis which is kind of the founding
document of ARS INF fur at the end of
that he kind of gestures to like other
problems now that we want to be able to
attack and one of them was this issue of
model

uh structure learning learning the
actual structure of the the the model in
that case it's a graph it's very high
problem as far as I understand and no
one really knows how to do it
yet yeah like a few things so you could
have an explicit distribution over
like I don't know how many lights are
going to be in the
room I think it could be from 1 to 10
and then you're then you could have
methods that oh it's seven now I'm going
to spin up seven a matrices for the
seven

lights but then that kind of like
question of unknown unknowns and being
able to in a potentially open-ended way
incorporate like new new rows in the
variables you already have like oh you
could go before you could go up down
left right now you can also go a
diagonal

Direction so that kind of like expanding
variables you have and then also like oh
now you have a new type of affordance or
you have a new sensory modality that

um at at least in terms of the examples
we have in open question um another
quote that I think is kind of just a
plant a footnote this is from
uh Toby St Clair
smi's uh
thesis he differentiates between the
structured learning and the learning
structure so learning
structure is about the former extending
the process of learning to a structured
setting whereas the latter learning the
underlying structures
themselves it's a kind of playful but
interesting question like are we
learning the structure of the patterns
in the
world with a fixed structure
internally that's one side of the coin
and the other side would be given the
structure of the patterns in the world
how can the actim model potentially
modify its own structure to accommodate
those patterns more
easily RX infer hopefully provides the
sandbox to experiment with those things
because of the possible open-endedness
of being able to combine different
nodes um I don't think any of our
examples from kass's work or from the
reactive Bas
team they but Fraser rovas correct me if
if this is not the case but like they
have a fixed
topology in
runtime it's not that the graph has a
certain connectedness and then something
occurs like a policy decision or some

kind of signal and then the graph
changes but and then like Dave mentioned
genetic algorithms
like there there's so many other pieces
like the parameter sweep mentioned
earlier but like parameter sweeping
isn't necessarily part of active
inference it's not part of the
particular partition it's not the
physics of action and perception it's
just a computational science method so
in practice a variety of methods come
into
play ranging from parameter sweeps
parallelization AB
testing so those all come into play to
kind of
tune or or give a bigger view on the
active inference generative model so
sometimes you are just interested in
building one generative
model other times it's like even for
just for fun and um speculative realism
even then you still may want to generate
a sequence of generative
models um just like if you were doing
linear aggression you would make from
simple on through like you have a um a
family from the most possible
interactions and including all the
variables to like the simplest model and
then there' be the combinations in in
between so there's something kind of
analogous to to to cross calibrating the
performance of different models that
have that have different structural
elements now there was a presentation uh
might have been a guest stream and I

suspect it was in 2020 where somebody
was addressing these using a formal
setting uh where
graphs combine and
split in response to particular kinds of
failures do you remember that an
ecological treatment
of
graphs I I do remember Bert D of the
bias lab saying that this idea of uh 40
40 St Factor graphs they allow for a
relatively robust model in the sense
that you can take out nodes and you know
the reactive message passing will just
kind of self-organize around that um I
don't know if that's actually been
demonstrated yet with with ar in
particular I think maybe that's more of
a theoretical um benefit so far uh but
yeah I'm not aware of actual examples of
that just yet there there's several
other um
I think it was the
supervised there's several other
approaches to to structure learning so
first off again a Toby paper this is
more on the category Theory here um this
is in terms of structured learning
because it describes agents with
structured interfaces for learning like
computer apis so that's super exciting
but now on the learning structure
side um there's earlier work
2020 Smith at all and then there's a
more recent work um which is quite
interesting
here okay the the notation and the
letters get a little funny and

interesting because like here the transition probabilities are encoded in C

but in the textbook we're just discussing like C is the preference on observations but here it's not so it just you know see that

um what happens in this paper is

uh it's in the context of amnest handwriting data and there are basically these um

there's an unknown unknown and not preset number of styles and then there's this uh question which is how many styles should we given our task is to categorize these digits into 0o through

nine how many styles should we spin up so in the extreme case there's only one style of zero one style of one one style of two that's just just like a big dis

big circle around the points in in in uh space on the Other Extreme like every single example is its own

style so both the one style and the every single one has its own style those are on two extremes of a structure

learning parameterization how many styles are there and so then the question is how do you find that sort of

prto optimal Bas optimal number of styles that actually provide you positive classification ability and so that's sometimes approach from this idea

of of on one

hand adding a latent factor and seeing if there's variability that that latent

factor can

explain and then testing um you you

using um BAS model selection okay here's

my model with n Styles here's n plus one

style is that extra style punching above

its weight or if it's diminishing

returns then the smaller model's better

so that's the kind of like add a latent

factor see if it helps and then move

forward whichever one's better and then

the other side of that is like the

pruning which is kind of like the sleep

or the synaptic pruning in that

situation you say okay given the N

parameters I have which one would be

like the first to go and then you remove

the least informative of the parameters

that you

have so that has led to some interesting

work probably more not open source than

open source on systems that are able to

in in different kinds of

ways add latent factors to explain and

explore deeper patterns in like

observation data or like in the

consequences of action and then

cyclically for example like in a sleep

wake wake type

situation add parameters during

wakefulness to explain more

variance remove parameters that are the

least informative during sleep to

promote model sparsity and

interpretability and simplicity

another thing just to note since the

question was like about training is

chapter nine is where the um the topic

of connecting to empirical data is

brought in

before chapter n it's like all kind of

like we proposed this model like we've

just a priori specified this model and

then we ran the model forward in a

generative direction to generate

synthetic

data the reverse

direction is let's just say we already

had a data set so we already have a data

set of people typing and eye movement or

of the mouse in the Maze or the particle

moving or whatever it is so we already

know what data we want to connect from

um so then we we still use the same

generative

model such that we we know that we want

these observations to connect up to the

kinds of observations here like we want

to have the location observation at

every time step so kind of you could

pre-register that model or you could

build it after the data were

collected um and then what's called

Model inversion is so here is

distribution of the

parameters given the observations and

actions and then that is being um

flipped it's proportional that's the

kind of infinity sign to here's the

probability of the parameters given the

model and then the probability of the

actions given the observations in the

model and so then this is

reconstructing latent parameters

Theta in this kind of General linear

regression

framework against the observations

conditioned upon the
observations so when we're generating
synthetic data you can think about that
as a distribution of
observations conditioned
upon
parameters like I'm just saying that the
room is um 30 degrees and now I'm going
to generate noisy thermometer data so
that's like generate me thermometer
observations conditioned upon the hidden
State

s the other way to do an inference the
inverted model would be tell me what you
can say about s the true temperature and
the room conditioned upon these
observations that I'm empirically
getting so those are kind of the two
directions of using a basian model and
um I think there's a lot of good work to
be done with making examples that do
that that kind of features just coming
online in
pmdp and it's there in RX and furb but
we don't have any larger scale or even
medium

examples but those would be some cool
things to look
into it it's it kind of returns to
chapter 6 like why are we building this
model like if we wanted to study
how are the relationships between
observations and hidden States learned
well then we would have a distribution
on a or we'd select from multiple a
matrices but once we kind of open up all
the possible directions of learning like
including preference learning like how

does the mouse know that it wants the
the food versus the aversive of stimulus
or how does it come to know the
consequences of its actions from a
movement perspective
perspective that that makes the space of
the model larger and larger and larger
combinatorically whereas the the case
where all the parameters are fixed so
there's like not really learning per se
there could still be real-time inference
but just it's it's not doing any deeper
learning that's like the that's like the
plug and chug special
case point
and then for every time you open up
learning on a given
variable um like first you have a
quantitative learning like just updating
the values in the cells then you can
think more of like structural learning
like the a matrix was 2×2 and now
it's going to be 2×3 or it's going to be
 3×3 and then you can get even into um
more kind of severe structure learning
like what about a second a matrix or
what about another modification on a
base graph each of those steps from
fixed quantitative tuning structural
tuning within a fix topology opening up
the topology that's open-ended each of
those could be like large increases in
the amount of sweeping that you have to
do
so if you want to make a playful example
with
open-endedness you can go in that
direction

um if you had more of
a uh if you had a question that defined
your pragmatic value such that you
wanted to get some utility out of your
model at some point you would probably
choose which distributions you want to
fix like I I'm just going to say it
prefers sugar and it avoids poison
because I'm
not asking the question how does it come
to prefer
sugar but then if you were asking how
does it come to prefer sugar you would
want to make that distribution
learnable because that's what would
address that question but if the
question was given that mice like sugar
how do they learn um associative um you
know between the color of the room and
the shock of the foot or something like
that like so the the
question from the outset sometimes
shapes what you situate as learnable or
not
yeah FR
sure I just I just wanted to say with
respect to the significance that that
discussion has to modeling uh a lot of
us you know I'm quite interested in in
you know building models of of of this
um the the relevance would be you know
you you kind of want to hardcode as much
as possible not worry about the topology
maybe initially not worry about the
multiplicity of the a matrices and just
kind of do the most uh truncated strict
beian inference you can on the most uh
you know cut down version of the problem

and then and then maybe iteratively up
um expose parts of the the model say
okay now now we've got our frog jumping
let's say or I think that classifies
frogs versus apples but you know what if
we need to say determine uh also what we
want to classify you know frogs versus
app versus oranges versus mice um how do
you determine that you want to even
classify frogs and apples that's that's
another action then you're GNA have to
build in so that that introduces a huge
amount of complexity so to to avoid that
if you're just interested in creating a
first step in a model the best thing to
do is just kind of ignore those problems
and hardcode it um essentially I was
just just thinking for for the relevance
that would have to to modeling efforts
yeah that's a great Point like it makes
me think about um here like we could
just draw a circle around each of these
bar charts and this is kind of like our
basian
graph kind of you know these are the
variables that we're considering it's
almost like that's an island coming out
of the ocean or or you know choose
choose a different way to think about it
but it's like if we wanted to explicitly
learn the prior we would have another
node which is our prior
on frogs and apples is it 0.
1.9 plus or minus .05 or is it 0.19 plus
orus
01 and then that would be the
learnability of this prior and then you
could say well how did it come to

believe that it was plus or minus 05 on
frogs and apples and then it's like so
the Buck has to stop somewhere in
practice for a particular model but
depending on what is useful or epistemic
for you you might pull back multiple
layers

Dave yeah there's actually quite a lot
of data

that's real world data that's the
analysis of which is can be relevant to
what we're talking about talking about
uh Behavior shaping from animal trainers
and then the formalization of that in
Skinner

style training

and uh bubbling up against what that
what what kinds of learning what kind of
teaching you actually can efficiently
explain is a nice a nice problem on
itself there certain you just you just
cannot readily explain in scenario set
up but there's a lot that you really
can yes so figure

91 where the experimenter's choice of
which stimuli to provide which test
questions to send etc those are
observations for the mouse and then what
the mouse does in terms of selected
actions that's the observations for the
experimenter this is kind of the cool
thing with with the basian methods is
you could pre-register or do a
statistical power

sweep what trains of experimental stimul
under the generative model as I
understand it what trains of
experimental stimuli would be

differentiable with to from each other
or or contrastable or not so like an
example of this that's that's very um
empirical is in fmri so the fmri signal
the Bold the brain oxygen related signal
that friston and and others have studied
with SPM for many years so that signal
happens on a time scale of several
seconds so they have a a graphical model
that includes the time scale of several
seconds and then they can simulate out
well if we had two stimuli and they were
alternating at the time scale of half a
second we wouldn't be able to dis to to
determine the the relative efficacy of
these two stimuli because they'd be too
convoluted by this like 3 second
response time on the other hand if you
had the the observations alternating
every
minute it would be too slow you'd only
get one change Point detection per
minute so then it's like well what time
scale should the observations be
alternating on to derive the highest
statistical
power then even with that calculation
there's further considerations like oh
but then if the the person cognitively
understands oh they're alternating every
16 seconds
so there's always many considerations in
practice um but that's kind of the the
fun of the iterated modeling is this
back and forth with the synthetic
propositions and the empirical
data if it's a situation where there is
empirical

data I mean the nice thing that it's
it's Beijing inference all the way down
on all of these things so we kind of
know what we're doing in terms of I mean
in some
sense um we're still doing be and
inference on model structure we're still
doing be and inference on you know the A
matrices and then versus the parameters
and so
on yes unified approach to um handling
and updating the variables whether
they're uh beliefs about State beliefs
about action beliefs about observations
Etc
um and and to kind of again bring it
back to the very
beginning we could be actually building
from the inside these active inference
models or like in figure
9.1 this could be a Transformer or a
neural
network so also these models could be
used to describe and map cognitive
architectures or systems are things that
aren't active inference things per
se but still have interactions across
interface
so sometimes you're actually building
it other times just observing it and
then building something to describe the
observations which would kind of be like
training a uh machine learning or
reinforcement learning model on language
or on something like that and and that
could be used to generate language so
again that has a lot in common with
other

methods it's just that this is a very
flexible phrasing of that
problem um
okay
well
so at least
this was sort of a
um General slm modeling
discussion but if there are topics that
people like want to explore I would
highly recommend slash ask put put the
questions on the page or in the table
before you can leave a comment to like
make sure that that they're
seen um and just kind of frontload what
you want to ask and that will help like
bring other people in and
and make make uh a fun learning
time okay any
um last
questions I I guess for chapter two
where uh
we're opening up for just any any uh
sailing discussions at all about any
parts of chapter 2 you know it's not
like it's gonna have to be a modeling
discussion or anything like that next
session yeah we can do any correct it'll
it'll just be we'll do two weeks a
chapter two and then we'll see if we
want to have another what we want to do
in in three weeks from
now cool cool okay I'll stop the
recording thank you